Table 1: Extracted substructures of DL models.

Models	Substructures	Category
LeNet-5 [1]	API Sequential Connection	Easy
AlexNet [2]	API Sequential Connection	Easy
VGGNet-16,19 [3]	API Sequential Connection	Easy
Wide ResNet [4]	Basic-Wide Block	Easy
	Wide-Dropout Block	Easy
MobileNet V1 [5]	Depthwise Separable Convolution	Easy
MobileNet V2 [6]	Inverted Residual Block	Easy
Xception [7]	Separable Convolution	Easy
SqueezeNeXt [8]	Two-stage Bottleneck Block with Separable Convolutions	Easy
SENet [9]	Squeeze-and-Excitation Block	Easy
GoogLeNet (Inception V1) [10]	Auxiliary Classifiers Block	Easy
Network in Network [11]	MLPconv Block	Easy
ResNet [12]	Bottleneck Block	Easy
	Residual Block	Medium
Multi-ResNet [13]	Multi-Residual Block	Medium
Res2Net [14]	Hierarchical Residual Block	Medium
RoR [15]	Multilevel Residual Block	Medium
DenseNet [16]	Dense Block	Medium
CFR-DenseNet [17]	Dense Block with Squeeze-and-Excitation Block	Medium
GoogLeNet (Inception V1) [10]	Multiple Parallel Convolution Branches	Medium
Inception V2 [18]	Inception V1 with Convolutions Factorization	Medium
Inception V3-V4 [19, 20]	Asymmetric Convolution Factorized Multi-Branching	Medium
ShuffleNet V1 [21]	Grouped Convolution Block	Medium
	Channel Shuffle Block	Medium
GCNet [22]	Global Context Block	Medium
SPPNet [23]	Spatial Pyramid Pooling Block	Medium
MixNet [24]	Mixed Depthwise Convolution Block	Medium
FCN [25]	Skip Architecture	Medium
ConvNeXt V1-V2 [26, 27]	Inverted Bottleneck Block	Medium
ScaleNet [28]	Scale Aggregation Block	Medium
CBAM [29]	CBAM Block	Medium
SqueezeNet [30]	Fire Block	Medium
ResNeXt [31]	Aggregated Residual Transformations Block	Hard
MobileNet V3 [32]	Inverted Residual Block with Squeeze-and-Excitation Block	Hard
ShuffleNet V1 [21]	Integrating Channel Shuffle Block of Residual Block	Hard
ShuffleNet V2 [33]	Channel Split Block followed Channel Shuffle Block	Hard
	Residual Block followed Channel Shuffle Block	Hard
Inception-ResNet [20]	Integrating Inception Block with Residual Block	Hard
SENet [9]	SE-Inception Block	Hard
	SE-ResNet Block	Hard
	SEResNeXt Block	Hard
	SEPreResNet Block	Hard
	SEPostResNet Block	Hard
	SE-Identity Block	Hard
SKNet [34]	Selective Kernel Block	Hard
IGCV V1-V3 [35, 36, 37]	Interleaved Group Convolutions Block	Hard
ScaleNet [28]	Integrating Scale Aggregation Block with Residual Block	Hard

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